SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Short Answers

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – Short Answers
Weightage:	30 % of CIA	**Total Marks:**	100 Marks
CO1 Statement:	*Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
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Section / Batch:	CSE	**Date:**	12-10-2026

Code of Conduct

I Trisha N 192311398 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- This test contains 50 questions, Total: 100 Marks.
- Duration: 1 hour.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Questions

Q1. A field engineer downloads a 15 MB firmware file on a mobile link that sustains an average of 12 Mbps. How long will the download take?

Q2. A background sync service on a smartphone draws an average current of 160 mA from a 4000 mAh battery. How many hours can the phone run this service before the battery is empty?

Q3. A handset sends a 1500-byte packet over a 2 Mbps uplink, and the propagation delay to the base station is 5 ms. What is the total one-way delay for the packet?

Q4. A mobile app compresses a 4 MB photo by 75% before uploading it over a 400 kbps packet-data link. How long does the upload take?

Q5. A planner locates co-channel cells by moving i = 2 cells along one axis and then j = 1 cell after a 60° turn. What cluster size N does this produce?

Q6. An operator adopts a 12-cell reuse cluster. What is the co-channel reuse ratio Q = D/R for this design?

Q7. Cells in a 7-cell cluster have a radius of 2 km. What is the minimum distance between the centres of two co-channel cells?

Q8. A district of 1000 sq. km is to be covered with hexagonal cells of 1 km radius. Approximately how many cells are needed?

Q9. Each cell must support 50 full-duplex channels, each using 25 kHz in each direction, with a 7-cell reuse pattern. How much total spectrum does the operator need?

Q10. A town is served by 10 cells. Traffic growth forces every cell to be split into microcells of half the original radius. How many cells will cover the same area after splitting?

Q11. A cell with 60 channels is converted to three 120° sectors, with channels shared equally. How many channels does each sector get?

Q12. A system has 1000 duplex channels and adopts a 4-cell cluster. How many channels does each cell receive?

Q13. A 7-cell reuse system operates where the path-loss exponent is 4. Considering only the six first-tier co-channel interferers, estimate the signal-to-interference ratio in dB.

Q14. A designer tests whether a 4-cell cluster can meet an 18 dB S/I target with a path-loss exponent of 4 and six first-tier interferers. What S/I does N = 4 actually give, and is the target met?

Q15. A drive test records a serving signal of -70 dBm, co-channel interference of -88 dBm and a noise floor of -100 dBm. Compute the SINR in dB.

Q16. A 1800 MHz carrier is used for a GSM rollout. What is its wavelength?

Q17. At 900 MHz, at what distance does the free-space path loss reach 120 dB?

Q18. The received power at a 1 km reference distance is -50 dBm in an urban area with a path-loss exponent of 3.5. What power is expected at 4 km?

Q19. A 2100 MHz receiver can tolerate a maximum Doppler shift of 150 Hz. What is the highest vehicle speed, in km/h, the link can support?

Q20. A mobile channel has a maximum Doppler shift of 200 Hz. Using T = 0.423 / f(d), estimate the channel's coherence time.

Q21. A GSM operator has 25 MHz in each direction, uses 200 kHz carriers and leaves one carrier as a guard. Each carrier carries 8 time slots. How many traffic channels are available?

Q22. A GSM burst period lasts 576.9 µs and the gross channel bit rate is 270.833 kbps. How many bits fit in one burst period?

Q23. An IS-95 CDMA link spreads a 9.6 kbps voice stream at a chip rate of 1.2288 Mcps. What is the processing gain in dB?

Q24. An FDMA system has 12.5 MHz of spectrum, 30 kHz channels and a 10 kHz guard band at each edge of the allocation. How many channels can it provide?

Q25. An LTE carrier has 1200 usable subcarriers spaced 15 kHz apart, grouped 12 per resource block. How many resource blocks does the carrier offer?

Q26. Handsets contend on a slotted-ALOHA random access channel with an offered load of G = 0.5 attempts per slot. What throughput does the channel achieve?

Q27. A location area has 25 cells, and the network must page 2000 incoming calls per hour in every cell of that area. How many paging messages are sent per hour?

Q28. A BSC controls 40 BTSs. Each BTS has 3 sectors, each sector has 2 TRXs, and each TRX carries 8 time slots. How many time slots does the BSC manage?

Q29. One MSC can serve 500,000 subscribers. How many MSCs does a city with 2.4 million subscribers need?

Q30. An Abis link runs on an E1 line of 2.048 Mbps, divided into 16 kbps sub-channels. How many sub-channels does the link carry?

Q31. An HLR stores 1.5 KB of profile data per subscriber. How much storage is needed for 8 million subscribers?

Q32. A VLR serves 60,000 visiting subscribers, and each performs 4 location updates per hour. How many location updates per second must the VLR process?

Q33. A GSM full-rate speech codec outputs 13 kbps, and channel coding raises this to 22.8 kbps over the air. What percentage overhead does channel coding add?

Q34. A 1400-byte voice payload is carried with a 20-byte IP header, an 8-byte UDP header and a 12-byte RTP header. What is the protocol efficiency?

Q35. An AMR codec produces a 244-bit speech frame every 20 ms, and each frame is sent with 40 bytes of IP/UDP/RTP headers. What bit rate does the stream use on the link?

Q36. In the packet core, a 1000-byte user packet is tunnelled with an 8-byte GTP-U header, an 8-byte UDP header and a 20-byte IP header. What percentage of the transmitted bytes is tunnelling overhead?

Q37. A GSM LAPDm signalling frame is 23 bytes long, of which 3 bytes are header. What is the frame's payload efficiency?

Q38. A PLMN uses a 15-digit IMSI with a 3-digit MCC and a 2-digit MNC. How many digits are left for the MSIN, and how many subscribers can the operator number?

Q39. Another operator in the same country is assigned a 3-digit MNC. How many unique subscriber identities can it issue?

Q40. A core-network packet gateway handles 40 Gbps. In the busy hour, each user averages 2 Mbps when active and is active 10% of the time. How many users can the gateway support?

Q41. On a data session, one-way delays are 10 ms on the radio link, 5 ms on backhaul, 3 ms in the core and 22 ms across the internet. What is the round-trip time?

Q42. A network starts handoff when the received signal falls to -95 dBm, and the minimum usable signal for voice is -100 dBm. What is the handoff margin?

Q43. A vehicle travels at 72 km/h along a highway lined with cells 2 km across. How many handoffs does it make in one hour?

Q44. A handoff margin of 6 dB is set. The signal fades by 0.1 dB per metre near the cell edge, and the user moves at 15 m/s. How much time does the network have to complete the handoff?

Q45. A serving cell is received at -85 dBm and a neighbour at -82 dBm, with a 4 dB handoff hysteresis. Will handoff be triggered? If not, how much stronger must the neighbour become?

Q46. One million inbound roamers each make 2 location updates per day. Each update exchanges 5 signalling messages of 200 bytes. How much roaming signalling traffic is generated per day?

Q47. A user crosses 12 cell boundaries per hour, and 20% of those crossings are also location-area boundaries. How many location updates does the user trigger per hour?

Q48. A cell serves 2000 subscribers. Each makes 3 calls in the busy hour, and calls last 2 minutes on average. What traffic, in Erlangs, does the cell carry?

Q49. Network statistics show 900 dropped calls out of 45,000 established calls in a day. What is the call drop rate?

Q50. A cell with 24 channels is offered 18 Erlangs at a 2% grade of service. How much traffic is actually carried, and what is the channel utilisation?

Q1. Download time of a 15 MB firmware file

Given: File size = 15 MB, Speed = 12 Mbps

Formula:

$$\text{Download time} = \text{Data size} / \text{Data rate}$$

Step 1: Convert MB into megabits.

$$15 \times 8 = 120 \text{ Mb}$$

Step 2: Calculate the time.

$$\text{T} = 120 / 12 = 10 \text{ seconds}$$

Answer: 10 seconds.

Q2. Battery life of a smartphone

Given: Battery capacity = 4000 mAh, Current = 160 mA

Formula:

$$\text{Battery life} = \text{Battery capacity} / \text{Current}$$

Substituting the values,

$$\text{T} = 4000 / 160 = 25 \text{ hours}$$

Answer: 25 hours.

Q3. Total one-way packet delay

Given: Packet size = 1500 bytes, Data rate = 2 Mbps, Propagation delay = 5 ms

Formula:

$$\text{T}_{\text{total}} = \text{T}_{\text{transmission}} + \text{T}_{\text{propagation}}$$

Step 1: Calculate transmission delay.

$$\text{T}_{\text{transmission}} = (1500 \times 8) / (2 \times 10^6) = 0.006 \text{ s} = 6 \text{ ms}$$

Step 2: Calculate total delay.

$$\text{T}_{\text{total}} = 6 + 5 = 11 \text{ ms}$$

Answer: 11 ms.

Q4. Upload time of a compressed photo

Given: Original size = 4 MB, Compression = 75%, Link speed = 400 kbps

Step 1: Calculate the compressed size.

$$\text{Size} = 4 \times (100 - 75) / 100 = 1 \text{ MB}$$

Step 2: Convert MB into kilobits.

$$1 \text{ MB} = 8000 \text{ kb}$$

Step 3: Calculate upload time.

$$\text{T} = 8000 / 400 = 20 \text{ seconds}$$

Answer: 20 seconds.

Q5. Cluster size using frequency reuse

Given: i = 2, j = 1

Formula:

$$N = i^2 + ij + j^2$$

Substituting the values,

$$N = 2^2 + (2 \times 1) + 1^2$$

$$N = 4 + 2 + 1 = 7$$

Answer: Cluster size = 7 cells.

Q6. Co-channel reuse ratio

Given: Cluster size N = 12

Formula:

$$Q = D / R = \sqrt{(3N)}$$

Substituting,

$$Q = \sqrt{(3 \times 12)} = \sqrt{36}$$

$$Q = 6$$

Answer: Co-channel reuse ratio = 6.

Q7. Minimum distance between co-channel cells

Given: Cell radius R = 2 km, Cluster size N = 7

Formula:

$$D = R\sqrt{(3N)}$$

Substituting,

$$D = 2\sqrt{(3 \times 7)} = 2\sqrt{21} = 2 \times 4.583$$

$$D = 9.17 \text{ km}$$

Answer: Minimum co-channel distance = 9.17 km.

Q8. Number of cells required for a district

Given: District area = 1000 km², Cell radius = 1 km

Formula for hexagonal cell area:

$$A = (3\sqrt{3} / 2)\, R^2$$

Substituting R = 1,

$$A = (3\sqrt{3} / 2) \times 1^2 \approx 2.598 \text{ km}^2$$

Number of cells required:

$$N = 1000 / 2.598 \approx 385$$

Answer: Approximately 385 cells.

Note: *This uses the standard hexagonal cell area formula. The earlier estimate of 1,155 cells used an incorrect area formula.*

Q9. Total spectrum required for frequency reuse

Given: 50 full-duplex channels per cell, Bandwidth = 25 kHz in each direction, Cluster size = 7

Formula:

$$B = 50 \times 25 \times 2 \times 7$$

Here, 2 accounts for the two directions of full-duplex communication.

$$B = 17500 \text{ kHz} = 17.5 \text{ MHz}$$

Answer: Total spectrum required = 17.5 MHz.

Q10. Cell splitting into microcells

Given: Original cells = 10, New radius = Half the original radius

Formula:

$$N_{new} = N_{old} \, (R_{old} / R_{new})^2$$

Substituting,

$$N_{new} = 10 \, (R / (R/2))^2 = 10 \times 2^2 = 40$$

Answer: 40 cells.

Q11. Channels per sector

Given: Total channels = 60, Number of sectors = 3

Formula:

$$\text{Channels per sector} = \text{Total channels} / \text{Number of sectors}$$

$$= 60 / 3 = 20$$

Answer: 20 channels per sector.

Q12. Channels per cell

Given: Total duplex channels = 1000, Cluster size = 4

Formula:

$$\text{Channels per cell} = \text{Total channels} / \text{Cluster size}$$

$$= 1000 / 4 = 250$$

Answer: 250 channels per cell.

Q13. Signal-to-interference ratio for a 7-cell cluster

Given: N = 7, n = 4, $i_0 = 6$

Formula:

$$S / I = (D/R)^n / i_0$$

Since D/R = √(3N),

$$S / I = (\sqrt{21})^4 / 6 = 21^2 / 6 = 441 / 6 = 73.5$$

Convert into dB:

$$(S/I)_{dB} = 10 \log_{10}(73.5) = 18.66 \text{ dB}$$

Answer: S/I = 18.66 dB.

Q14. S/I for a 4-cell cluster

Given: N = 4, n = 4, i_0 = 6, Target = 18 dB

Formula:

$$S / I = (\sqrt{(3N)})^n / i_0$$

Substituting,

$$S / I = (\sqrt{12})^4 / 6 = 144 / 6 = 24$$

Converting into dB,

$$10 \log_{10}(24) = 13.80 \text{ dB}$$

Since 13.80 < 18, the target is not achieved.

Answer: S/I = 13.80 dB; the target is not met.

Q15. Calculate SINR

Given: Signal power = −70 dBm, Interference power = −88 dBm, Noise power = −100 dBm

SINR is the ratio of signal power to the sum of interference and noise powers.

Step 1: Convert the powers to milliwatts.

$$S = 10^{-70/10} = 10^{-7} \text{ mW}$$

$$I = 10^{-88/10} = 1.585 \times 10^{-9} \text{ mW}$$

$$N = 10^{-100/10} = 10^{-10} \text{ mW}$$

Step 2: Calculate SINR.

$$SINR = 10 \log_{10}[\ 10^{-7} / (1.585 \times 10^{-9} + 10^{-10})\ ]$$

$$SINR \approx 17.73 \text{ dB}$$

Answer: SINR ≈ 17.73 dB.

Q16. Wavelength of a GSM carrier

Given: Frequency f = 1800 MHz, Speed of light c = 3 × 10^{8} m/s

Formula:

$$\lambda = c / f$$

$$\lambda = (3 \times 10^8) / (1800 \times 10^6) = 0.1667 \text{ m}$$

Answer: Wavelength = 0.167 m or 16.7 cm.

Q17. Distance for 120 dB free-space path loss

Given: Frequency = 900 MHz, FSPL = 120 dB

Formula:

$$\mathrm{FSPL} = 32.44 + 20\log_{10}(\mathrm{f}) + 20\log_{10}(\mathrm{d})$$

Here, f is in MHz and d is in km.

$$120 = 32.44 + 20\log_{10}(900) + 20\log_{10}(\mathrm{d})$$

$$20\log_{10}(\mathrm{d}) = 120 - 32.44 - 59.08 = 28.48$$

$$\mathrm{d} = 10^{28.48/20} \approx 26.5\ \mathrm{km}$$

Answer: Approximately 26.5 km.

Note: *The earlier value of 265 km was a calculation error.*

Q18. Received power at 4 km

Given: Reference power = −50 dBm at 1 km, Path-loss exponent n = 3.5, Distance = 4 km

Formula:

$$P_r(\mathrm{d}) = P_r(\mathrm{d}_0) - 10\mathrm{n}\log_{10}(\mathrm{d}\ /\ \mathrm{d}_0)$$

Substituting,

$$P_r(4) = -50 - 10(3.5)\log_{10}(4/1)$$

$$= -50 - 35(0.6021) = -50 - 21.07 = -71.07\ \mathrm{dBm}$$

Answer: −71.1 dBm.

Q19. Maximum vehicle speed from Doppler shift

Given: Carrier frequency = 2100 MHz, Maximum Doppler shift = 150 Hz

Formula:

$$f_d = v\ f_c\ /\ c$$

Rearranging,

$$v = f_d\ c\ /\ f_c$$

$$v = (150 \times 3 \times 10^8)\ /\ (2.1 \times 10^9) = 21.43\ \mathrm{m/s}$$

Convert to km/h:

$$v = 21.43 \times 3.6 = 77.14\ \mathrm{km/h}$$

Answer: Approximately 77.1 km/h.

Q20. Coherence time of a mobile channel

Given: Maximum Doppler shift = 200 Hz

Formula:

$$T_c = 0.423\ /\ f_d$$

Substituting,

$$T_c = 0.423 / 200 = 0.002115 \text{ s}$$

Converting into milliseconds,

$$T_c = 2.115 \text{ ms}$$

Answer: Coherence time = 2.115 ms.

Q21. Number of GSM traffic channels

Given: Spectrum = 25 MHz in each direction, Carrier bandwidth = 200 kHz, One guard carrier, 8 time slots per carrier

Step 1: Calculate the number of carriers.

$$N = 25 / 0.2 - 1 = 124$$

Step 2: Calculate traffic channels.

$$124 \times 8 = 992$$

Answer: 992 traffic channels.

Q22. Bits in one GSM burst period

Given: Burst period = 576.9 µs, Bit rate = 270.833 kbps

Formula:

$$\text{Bits} = \text{Bit rate} \times \text{Time}$$

Convert the units:

$$576.9 \text{ µs} = 576.9 \times 10^{-6} \text{ s}$$

Then,

$$\text{Bits} = 270833 \times 576.9 \times 10^{-6} \approx 156.25 \text{ bits}$$

Answer: Approximately 156 bits per burst period.

Q23. Processing gain of an IS-95 CDMA link

Given: Data rate = 9.6 kbps, Chip rate = 1.2288 Mcps

Formula:

$$G_p = \text{Chip rate} / \text{Data rate}$$

$$G_p = (1.2288 \times 10^6) / (9.6 \times 10^3) = 128$$

Convert into dB:

$$G_p(\text{dB}) = 10 \log_{10}(128) = 21.07 \text{ dB}$$

Answer: Processing gain = 21.1 dB.

Q24. Number of FDMA channels

Given: Total bandwidth = 12.5 MHz, Channel bandwidth = 30 kHz, Edge guard band = 10 kHz at each edge

Step 1: Calculate usable bandwidth.

$$B = 12.5 - 0.01 - 0.01 = 12.48 \text{ MHz}$$

Step 2: Calculate the number of channels.

$$N = (12.48 \times 1000) / 30 = 416$$

Answer: 416 channels.

Q25. LTE resource blocks

Given: Usable subcarriers = 1200, Subcarriers per resource block = 12

Formula:

$$N_{RB} = \text{Total subcarriers} / \text{Subcarriers per RB}$$

$$N_{RB} = 1200 / 12 = 100$$

Answer: 100 resource blocks.

Q26. Slotted-ALOHA throughput

Given: Offered load G = 0.5 attempts per slot

Formula:

$$S = G\, e^{-G}$$

Substituting,

$$S = 0.5\, e^{-0.5}$$

Since $e^{-0.5} \approx 0.6065$,

$$S = 0.5 \times 0.6065 = 0.3033$$

Answer: Throughput = 0.303 packets per slot.

Q27. Paging messages per hour

Given: Number of cells = 25, Calls per cell per hour = 2000

Formula:

$$\text{Total paging messages} = \text{Number of cells} \times \text{Messages per cell}$$

$$= 25 \times 2000 = 50{,}000$$

Answer: 50,000 paging messages per hour.

Q28. Total time slots managed by the BSC

Given: 40 BTSs, 3 sectors per BTS, 2 TRXs per sector, 8 time slots per TRX

Formula:

$$N = 40 \times 3 \times 2 \times 8$$

$$N = 1920$$

Answer: 1920 time slots.

Q29. Number of MSCs required

Given: City population of subscribers = 2.4 million, Capacity per MSC = 500,000 subscribers

Formula:

$$N = \text{Total subscribers} / \text{Capacity per MSC}$$

$$N = 2{,}400{,}000 / 500{,}000 = 4.8$$

Since the number of MSCs must be a whole number, round up.

Answer: 5 MSCs.

Q30. Number of sub-channels on an E1 link

Given: E1 data rate = 2.048 Mbps, Sub-channel rate = 16 kbps

Formula:

$$N = \text{Total data rate} / \text{Sub-channel rate}$$

$$N = 2048 / 16 = 128$$

Answer: 128 sub-channels.

Q31. HLR storage requirement

Given: Data per subscriber = 1.5 KB, Subscribers = 8 million

Formula:

$$\text{Storage} = \text{Data per subscriber} \times \text{Number of subscribers}$$

$$= 1.5 \times 8{,}000{,}000 = 12{,}000{,}000 \text{ KB}$$

Using decimal units, this is approximately 12 GB.

Answer: Approximately 12 GB.

Q32. VLR location updates per second

Given: Visiting subscribers = 60,000, Updates per subscriber per hour = 4

Step 1: Calculate hourly updates.

$$60{,}000 \times 4 = 240{,}000$$

Step 2: Convert to updates per second.

$$N = 240{,}000 / 3600 = 66.67$$

Answer: Approximately 66.67 updates per second.

Q33. Channel-coding overhead

Given: Original speech rate = 13 kbps, Coded rate = 22.8 kbps

Formula:

$$\text{Overhead} = (\text{Coded rate} - \text{Original rate}) \,/\, \text{Original rate} \times 100$$

$$= (22.8 - 13) \,/\, 13 \times 100 = 9.8 \,/\, 13 \times 100 = 75.38\%$$

Answer: Channel-coding overhead = 75.38%.

Q34. Protocol efficiency of a voice packet

Given: Payload = 1400 bytes, IP header = 20 bytes, UDP header = 8 bytes, RTP header = 12 bytes

Step 1: Calculate total packet size.

$$\mathrm{L} = 1400 + 20 + 8 + 12 = 1440 \text{ bytes}$$

Step 2: Calculate protocol efficiency.

$$\eta = \text{Payload} \,/\, \text{Total packet size} \times 100$$

$$\eta = 1400 \,/\, 1440 \times 100 = 97.22\%$$

Answer: Protocol efficiency = 97.22%.

Q35. Bit rate of an AMR voice stream

Given: Speech frame = 244 bits every 20 ms, Headers = 40 bytes per frame

Step 1: Convert header bytes to bits.

$$40 \times 8 = 320 \text{ bits}$$

Step 2: Calculate total bits per frame.

$$\mathrm{L} = 244 + 320 = 564 \text{ bits}$$

Step 3: Calculate bit rate.

$$\mathrm{R} = 564 \,/\, (20 \times 10^{-3}) = 28{,}200 \text{ bps} = 28.2 \text{ kbps}$$

Answer: 28.2 kbps.

Q36. Tunnelling overhead in the packet core

Given: User packet = 1000 bytes, GTP-U header = 8 bytes, UDP header = 8 bytes, IP header = 20 bytes

Step 1: Calculate total overhead.

$$\mathrm{H} = 8 + 8 + 20 = 36 \text{ bytes}$$

Step 2: Calculate total transmitted size.

$$\mathrm{L} = 1000 + 36 = 1036 \text{ bytes}$$

Step 3: Calculate overhead percentage.

$$\text{Overhead} = 36 \,/\, 1036 \times 100 = 3.47\%$$

Answer: Tunnelling overhead = 3.47%.

Q37. Payload efficiency of a GSM LAPDm frame

Given: Frame size = 23 bytes, Header = 3 bytes

Step 1: Calculate payload size.

$$\mathrm{P} = 23 - 3 = 20 \text{ bytes}$$

Step 2: Calculate payload efficiency.

$$\eta = 20 \,/\, 23 \times 100 = 86.96\%$$

Answer: Payload efficiency = 86.96%.

Q38. Subscriber identities with a 2-digit MNC

Given: IMSI length = 15 digits, MCC = 3 digits, MNC = 2 digits

Step 1: Calculate the MSIN length.

$$\text{MSIN} = 15 - 3 - 2 = 10 \text{ digits}$$

Step 2: Calculate the possible MSIN combinations.

$$\mathrm{N} = 10^{10}$$

Answer: 10 MSIN digits, allowing up to 10 billion identities per assigned MCC/MNC combination.

Q39. Subscriber identities with a 3-digit MNC

Given: IMSI length = 15 digits, MCC = 3 digits, MNC = 3 digits

Step 1: Calculate the MSIN length.

$$\text{MSIN} = 15 - 3 - 3 = 9 \text{ digits}$$

Step 2: Calculate the possible combinations.

$$\mathrm{N} = 10^{9}$$

Answer: 9 MSIN digits, allowing up to 1 billion identities per assigned MCC/MNC combination.

Q40. Number of users supported by a packet gateway

Given: Gateway capacity = 40 Gbps, Active user rate = 2 Mbps, Activity factor = 10%

Step 1: Convert the gateway capacity.

$$40 \text{ Gbps} = 40{,}000 \text{ Mbps}$$

Step 2: Calculate the number of simultaneously active users.

$$\mathrm{N_a} = 40{,}000 \,/\, 2 = 20{,}000$$

Step 3: Calculate the total number of users.

$$\mathrm{N} = 20{,}000 \,/\, 0.10 = 200{,}000$$

Answer: 200,000 users.

Q41. Round-trip time of a data session

Given: Radio delay = 10 ms, Backhaul = 5 ms, Core = 3 ms, Internet = 22 ms

Step 1: Calculate one-way delay.

$$T = 10 + 5 + 3 + 22 = 40 \text{ ms}$$

Step 2: Calculate round-trip time.

$$RTT = 2 \times 40 = 80 \text{ ms}$$

Answer: RTT = 80 ms.

Q42. Handoff margin

Given: Handoff threshold = −95 dBm, Minimum usable signal = −100 dBm

Formula:

$$M = P_{handoff} - P_{minimum}$$

$$M = -95 - (-100) = 5 \text{ dB}$$

Answer: Handoff margin = 5 dB.

Q43. Number of handoffs in one hour

Given: Vehicle speed = 72 km/h, Cell width = 2 km

Formula:

$$\text{Handoffs per hour} = \text{Distance travelled per hour} / \text{Cell width}$$

$$N = 72 / 2 = 36$$

Answer: 36 handoffs per hour.

Q44. Time available to complete a handoff

Given: Handoff margin = 6 dB, Fade rate = 0.1 dB/m, Speed = 15 m/s

Step 1: Calculate the distance corresponding to the margin.

$$d = 6 / 0.1 = 60 \text{ m}$$

Step 2: Calculate the available time.

$$T = d / v = 60 / 15 = 4 \text{ s}$$

Answer: 4 seconds.

Q45. Handoff decision using hysteresis

Given: Serving signal = −85 dBm, Neighbour signal = −82 dBm, Hysteresis = 4 dB

Step 1: Calculate the neighbour's signal advantage.

$$\Delta P = -82 - (-85) = 3 \text{ dB}$$

Step 2: Compare with the hysteresis value.

$$3 \text{ dB} < 4 \text{ dB}$$

The handoff condition is not satisfied. The neighbour must improve by another 1 dB.

Answer: Handoff is not triggered; the neighbour must become 1 dB stronger.

Q46. Roaming signalling traffic per day

Given: Inbound roamers = 1 million, Updates per day = 2 per user, Messages per update = 5, Message size = 200 bytes

Step 1: Calculate the total messages.

$$N = 1{,}000{,}000 \times 2 \times 5 = 10{,}000{,}000 \text{ messages}$$

Step 2: Calculate the total data.

$$D = 10{,}000{,}000 \times 200 = 2{,}000{,}000{,}000 \text{ bytes}$$

Answer: 2 GB per day, excluding additional protocol overhead.

Q47. Location updates due to location-area crossings

Given: Cell crossings = 12 per hour, Location-area boundaries = 20%

Formula:

$$\text{Location updates} = 12 \times 20 \,/\, 100$$

$$= 12 \times 0.2 = 2.4$$

Answer: 2.4 location updates per hour on average.

Q48. Traffic carried by a cell

Given: Subscribers = 2000, Calls per subscriber = 3 in the busy hour, Average call duration = 2 minutes

Formula:

$$A = (\text{Number of calls} \times \text{Average duration in minutes}) \,/\, 60$$

Step 1: Calculate the number of calls.

$$2000 \times 3 = 6000 \text{ calls}$$

Step 2: Calculate traffic in Erlangs.

$$A = (6000 \times 2) \,/\, 60 = 200$$

Answer: 200 Erlangs.

Q49. Call drop rate

Given: Dropped calls = 900, Established calls = 45,000

Formula:

$$\text{Call drop rate} = \text{Dropped calls} / \text{Established calls} \times 100$$

$$= 900 / 45{,}000 \times 100 = 2\%$$

Answer: Call drop rate = 2%.

Q50. Traffic carried and channel utilisation

Given: Number of channels = 24, Offered traffic = 18 Erlangs, Grade of service = 2%

Using the Erlang B formula,

$$B(C, A) = (A^C / C!) / \Sigma_{k=0}{}^{C} (A^k / k!)$$

For C = 24 and A = 18, the blocking probability is approximately 2%. Therefore, the carried traffic is:

$$A_c = A(1 - B) = 18(1 - 0.02) = 17.64 \text{ Erlangs}$$

Channel utilisation is:

$$U = 17.64 / 24 \times 100 = 73.5\%$$

Answer: Approximately 17.64 Erlangs carried and 73.5% channel utilisation.

Note: *The exact Erlang B result should be used if your instructor expects a table or calculator-based value; 17.64 Erlangs is the approximation using the stated 2% blocking probability.*